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Bachelor of Liberal Arts & Sciences

SSCMATH303 Mathematical Statistics (SPRING 2025)

Project

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Date of submission: 17 of May 2025

Reference system: APA

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1 Introduction

This paper examines algorithms of sampling different most common distributions via different sampling methods. The paper is divided in two main sections: Theory and Experiments.

In the theory part three major sampling ideas are discussed, while in the Experiments part those ideas are applied to actually get a sample from various distributions.
The key assumption made in this paper is the following: We are able to sample from a Uniform$(0, 1)$ and we can sample random positive integers.

2 Theory

2.1 Inverse Transform Sampling Method

Inverse transform sampling is a general procedure for generating samples from a continuous distribution, provided its cumulative distribution function (CDF) is known and invertible. Let X be a continuous random variable with CDF

$$F_X(x) \;=\; \Pr[X \le x],$$

and density $f_X(x) = F'_X(x)$. If $U \sim \mathsf{Uniform}(0, 1)$, then the transformation

$$X = F_X^{-1}(U)$$

yields a random variable with the desired distribution. Indeed, for any x,

$$\Pr\big(F_X^{-1}(U) \le x\big) = \Pr\big(U \le F_X(x)\big) = F_X(x),$$

which coincides with the original CDF.

Algorithm:

1. Generate $U \sim \mathsf{Uniform}(0, 1)$.
2. Compute $X = F_X^{-1}(U)$.

2.2 Accept–Reject Sampling Method

Accept–reject sampling (also called rejection sampling) is a general technique for drawing independent samples from a target density $f_X(x)$, provided there exists an envelope (proposal)

density $g(x)$ and a constant $M \geq 1$ such that

$$f_X(x) \;\leq\; M\, g(x) \quad \forall\, x.$$

The method proceeds by simulating candidate points from $g(x)$ and accepting them with probability proportional to the ratio f_X/g. Concretely, let $U_1, U_2 \sim \mathsf{Uniform}(0,1)$ and let $Y \sim g$. Then

$$\Pr\big(\text{accept } Y\big) = \int \frac{f_X(y)}{M\, g(y)}\, g(y)\, dy = \frac{1}{M} \int f_X(y)\, dy = \frac{1}{M},$$

so that the overall acceptance rate is $1/M$.

Algorithm:

1. Draw $Y \sim g(y)$ and $U \sim \mathsf{Uniform}(0,1)$ independently.
2. If
$$U \;\leq\; \frac{f_X(Y)}{M\, g(Y)},$$
accept $X = Y$; otherwise, reject Y and return to step 1.

To verify correctness, note that the joint density of an accepted pair (Y, U) restricted to the acceptance region is

$$g(y)\, \mathbf{1}\Big\{u \leq \frac{f_X(y)}{M\, g(y)}\Big\}$$

and hence the marginal density of accepted Y is

$$\int_0^{f_X(y)/(M\, g(y))} g(y)\, du = \frac{f_X(y)}{M}\,,$$

which, when renormalized by the acceptance probability $1/M$, recovers $f_X(y)$.

2.3 Sampling from a Joint PDF via Gibbs Sampling

Let $f_{X,Y}(x, y)$ be the target joint probability density on $\mathbb{R}^2$.

Gibbs Sampling. When the full conditional densities

$$f_{X|Y}(x \mid y) \quad \text{and} \quad f_{Y|X}(y \mid x)$$

are available in closed form, one can construct a Markov chain whose stationary distribution is $f_{X,Y}$. Starting from an initial $(x^{(0)}, y^{(0)})$, iterate for $t = 1, 2, \ldots$:

1. Draw $x^{(t)} \sim f_{X|Y}\big(x \mid y^{(t-1)}\big)$.
2. Draw $y^{(t)} \sim f_{Y|X}\big(y \mid x^{(t)}\big)$.

Under mild regularity conditions, $\{(x^{(t)}, y^{(t)})\}$ converges in distribution to the target $f_{X,Y}$. Gibbs sampling scales well to higher dimensions but may exhibit slow mixing if the components are highly correlated.

3 Experiments

3.1 Bernoulli

Let $U \sim \mathsf{Uniform}(0, 1)$. The Bernoulli($p$) distribution has CDF

$$F(x) = \begin{cases} 0, & x < 0, \\ 1 - p, & 0 \le x < 1, \\ 1, & x \ge 1. \end{cases}$$

The inverse CDF is

$$F^{-1}(u) = \begin{cases} 0, & 0 \le u < 1 - p, \\ 1, & 1 - p \le u \le 1. \end{cases}$$

Thus, to generate $X \sim \text{Bernoulli}(p)$ we set

$$X = F^{-1}(U) = \begin{cases} 1, & U \ge 1 - p, \\ 0, & U < 1 - p, \end{cases}$$

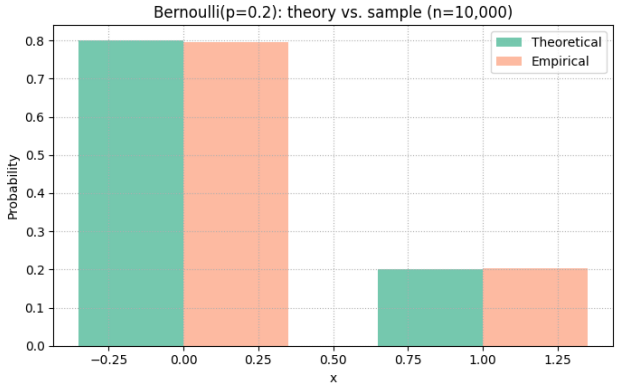


Figure 1: Bernoulli real vs Bernoulli sampled

3.2 Binomial

We sample $X \sim \text{Binomial}(n, p)$ by summing n independent Bernoulli(p) draws:

$$B_i \sim \text{Bernoulli}(p), \quad X = \sum_{i=1}^{n} B_i.$$

In practice, generate $U_{ij} \sim U(0, 1)$ and set

$$B_i = \mathbf{1}\{U_{ij} < p\}, \quad X = \sum_{i=1}^{n} B_i.$$

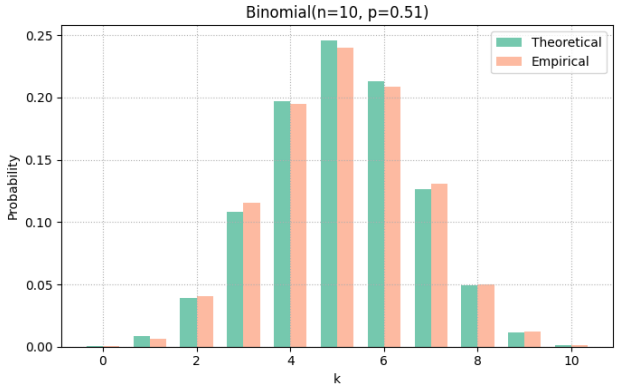


Figure 2: Binomial ($p = 3$, $n = 10$)

3.3 Geometric

Target distribution: Let $X \sim \text{Geometric}(p)$, the number of failures before the first success, with

$$\Pr(X = k) = (1-p)^k\, p, \quad k = 0, 1, 2, \ldots.$$

Method: Sequential Bernoulli (Waiting-Time) We leverage the fact that a Geometric random variable counts how many independent Bernoulli(p) trials (failures) occur before the first success. Given a pre-sampled Bernoulli matrix, we extract the waiting time:

$$X = \min\{\, i \geq 0 : B_i = 1\} \;-\; 0 \quad \text{with } B_i \overset{\text{iid}}{\sim} \text{Bernoulli}(p).$$

Algorithm:

1. Generate a matrix $\{B_{j,i}\}$ of size $n_{\text{samples}} \times M$, where $B_{j,i} \sim \text{Bernoulli}(p)$.
2. For each sample j, find the first index i such that $B_{j,i} = 1$.
3. Set $X_j = i$. This is the count of failures before the first success.

Since the Bernoulli trials are independent,

$$\Pr(X = k) \;=\; \Pr(B_0 = 0, \ldots, B_{k-1} = 0, B_k = 1) = (1-p)^k\, p,$$

matching the Geometric(p) PMF exactly.

3.4 Negative Binomial

Method: Sequential Bernoulli Sampling for NegBinomial(r, p)

Let $\{B_i\}_{i \geq 1}$ be i.i.d. Bernoulli(p) trials. Define

$$T = \min\Big\{n : \sum_{i=1}^{n} B_i = r\Big\}, \quad X = T - r.$$

Then X counts the number of failures (zeros) observed before the rth success. Since $\sum_{i=1}^{n} B_i$ increments by one on each success, the stopping time T exactly captures when the rth success occurs. The distribution of X is

$$P(X = k) = P\big(T = k + r\big) = \binom{k + r - 1}{k}(1 - p)^k \, p^r, \quad k = 0, 1, 2, \ldots$$

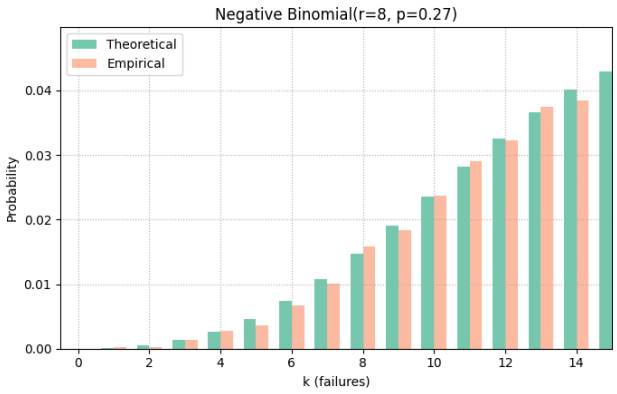


Figure 3: Negative binomial empirical sample vs real distribution

3.5 Hypergeometric

Method: Direct Sampling via Random Permutation

Consider a population of size N, with K "successes" and $N - K$ "failures." The Hypergeometric distribution $\mathrm{HG}(N, K, n)$ gives the probability of drawing X successes in n draws without replacement.

$$P(X = k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}, \quad k = 0, 1, \ldots, n.$$

Algorithm:

1. Generate a random permutation π of $\{1, 2, \ldots, N\}$ (each permutation with equal probability).
2. Let $A = \{\pi(1), \pi(2), \ldots, \pi(n)\}$ be the first n selected positions.

3. Return $X = |\, A \cap \{1, 2, \ldots, K\}\,|$, the count of selected indices in the "success" range.

Since all subsets of size n are equally likely under a random permutation, this produces exactly $\mathrm{HG}(N, K, n)$.

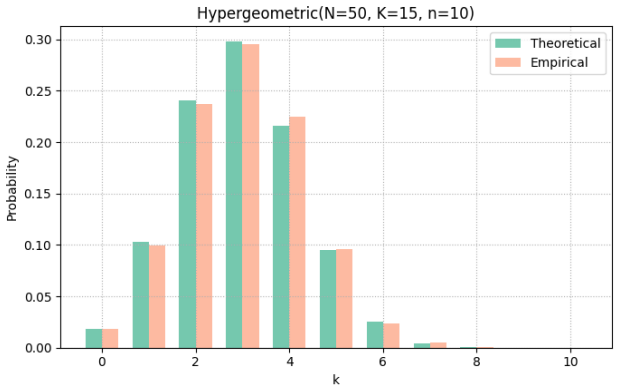


Figure 4: Enter Caption

3.6 Poisson

Method: Poisson Sampling via Exponential Interarrival Times

A Poisson process with rate $\lambda > 0$ has interarrival times $Y_i \sim \text{Exponential}(\lambda)$ i.i.d. The count of events in a fixed interval $[0, T]$ is

$$N(T) = \max\{n : S_n \leq T\}, \quad S_n = \sum_{i=1}^{n} Y_i.$$

One shows that $N(T) \sim \text{Poisson}(\lambda T)$.

Algorithm:

1. Initialize $t = 0$, $n = 0$.
2. **While** $t < T$:
   (a) Draw $Y \sim \text{Exponential}(\lambda)$.
   (b) Set $t \leftarrow t + Y$.

(c) **If** $t < T$, increment $n \leftarrow n + 1$.

3. Return $N = n$.

Thus by summing exponentials until the total exceeds T, the number of accepted increments is exactly a Poisson(λT) random draw. The empirical histogram of these counts matches the theoretical PMF

$$P(N = k) = \frac{(\lambda T)^k}{k!} e^{-\lambda T}, \quad k = 0, 1, 2, \ldots$$

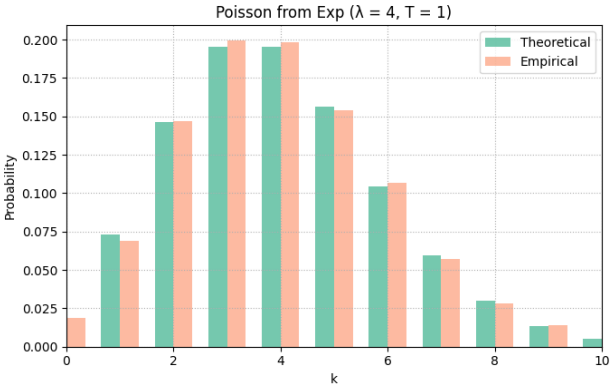


Figure 5

3.7 Uniform

Method: Inverse Transform Sampling for Uniform$[a, b]$

For $X \sim \text{Uniform}(a, b)$, the CDF is

$$F(x) = P(X \leq x) = \begin{cases} 0, & x < a, \\ \frac{x-a}{b-a}, & a \leq x \leq b, \\ 1, & x > b. \end{cases}$$

Algorithm:

1. Generate $U \sim \text{Uniform}(0, 1)$.

2. Compute

$$X = F^{-1}(U) = a + (b - a)\, U.$$

Since U is uniform on $[0, 1]$, X is uniform on $[a, b]$.

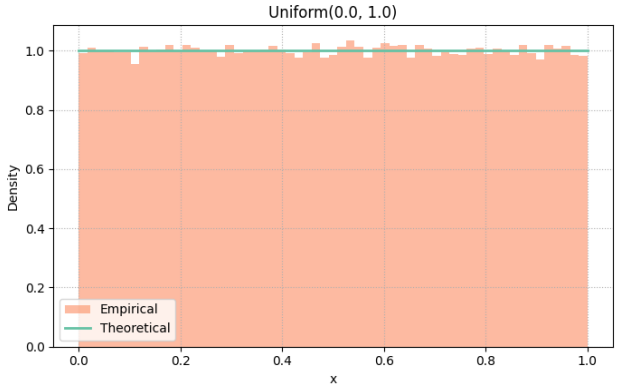


Figure 6: Enter Caption

3.8 Exponential

Method: Inverse Transform Sampling for Exponential(λ)

Let X have the Exponential distribution with rate parameter $\lambda > 0$. Its CDF is

$$F(x) = P(X \le x) = \begin{cases} 1 - e^{-\lambda x}, & x \ge 0, \\ 0, & x < 0. \end{cases}$$

Algorithm:

1. Draw $U \sim \text{Uniform}(0, 1)$.

2. Solve $F(x) = U$ for x. That is,

$$1 - e^{-\lambda x} = U \implies e^{-\lambda x} = 1 - U \implies x = -\frac{1}{\lambda} \ln\big(1 - U\big).$$

3. Return $X = -\frac{1}{\lambda} \ln\big(1 - U\big)$.

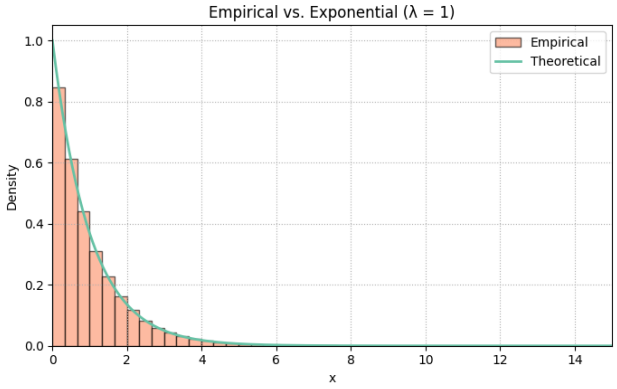


Figure 7: Exponential ($\lambda = 1$) Empirical vs. real

3.9 Gamma

Method: Sum of Exponentials (Erlang) for Gamma(r, λ)

When r is a positive integer, the Gamma distribution with shape r and rate λ can be represented as the sum of r i.i.d. Exponential(λ) random variables:

$$X = \sum_{i=1}^{r} Y_i, \quad Y_i \overset{\text{i.i.d.}}{\sim} \text{Exponential}(\lambda).$$

Since each Y_i has PDF $f(y) = \lambda e^{-\lambda y}$ for $y \geq 0$, the sum of r such exponentials yields the Gamma PDF:

$$f_X(x) = \frac{\lambda^r x^{r-1} e^{-\lambda x}}{(r-1)!}, \quad x \geq 0.$$

Algorithm:

1. For each sample, draw r i.i.d. exponentials:

$$Y_i = -\frac{1}{\lambda} \ln(1 - U_i), \quad U_i \sim \text{Uniform}(0, 1), \quad i = 1, \ldots, r.$$

2. Compute $X = \sum_{i=1}^{r} Y_i$.

3. Return X.

Using this idea Gamma(r, λ) can be obtained for $\forall n \in \mathbb{N}$.

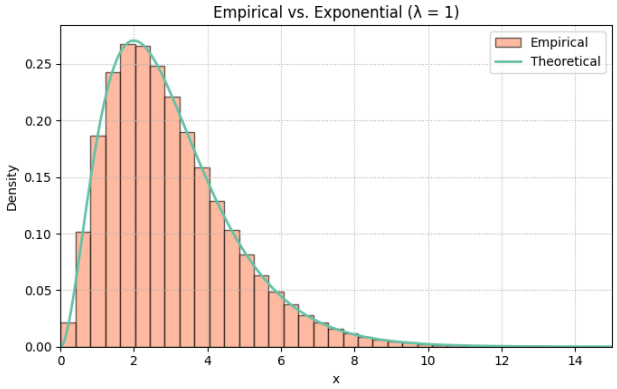


Figure 8: Empirical vs. Exp($\lambda = 1$)

Method: Accept–Reject Sampling for Gamma$(r_{\text{new}}, \lambda)$ with $n \notin \mathbb{N}$

Let the target density be the Gamma density with shape $r_{\text{new}} > 0$ and rate $\lambda > 0$,

$$f(x) = \frac{\lambda^{r_{\text{new}}}}{\Gamma(r_{\text{new}})} x^{r_{\text{new}}-1} e^{-\lambda x}, \quad x > 0.$$

Choose as proposal an Exponential density with rate $\lambda_{\text{prop}} > 0$,

$$g(x) = \lambda_{\text{prop}} e^{-\lambda_{\text{prop}} x}, \quad x > 0.$$

Define the likelihood ratio

$$R(x) = \frac{f(x)}{g(x)}$$

Compute its supremum (for loop for every value)

$$R^* = \sup_{x>0} R(x),$$

and then set

$$M \;=\; 1.05\, R(x^*) \quad \text{(a small safety margin)}.$$

Algorithm:

1. Precompute $M \geq R^*$ as above.

2. **Repeat** until you have n accepted draws:

   (a) Draw $X_{\text{prop}} \sim g(x)$ (i.e. $X_{\text{prop}} \sim \text{Exponential}(\lambda_{\text{prop}})$).

   (b) Draw $U \sim \text{Uniform}(0, 1)$.

   (c) **If** $U \leq \dfrac{f(X_{\text{prop}})}{M\, g(X_{\text{prop}})} = \dfrac{R(X_{\text{prop}})}{M}$, **accept** X_{prop}; **else** reject.

This procedure yields i.i.d. samples from the desired Gamma$(r_{\text{new}}, \lambda)$ law (Figure 9).

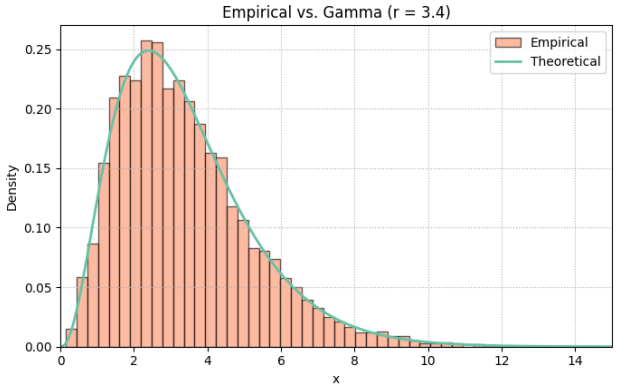


Figure 9: Empirical vs Gamma ($r = 3.4$)

3.10 Beta

Method: Gamma-Ratio Sampling for Beta(α, β)

The Beta distribution with parameters $\alpha > 0$, $\beta > 0$ has density

$$f_X(x) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\,\Gamma(\beta)}\, x^{\alpha-1}\,(1-x)^{\beta-1}, \quad 0 < x < 1.$$

One can generate a Beta(α, β) random variate by exploiting the property that if

$$U \sim \Gamma(\alpha, 1), \qquad V \sim \Gamma(\beta, 1),$$

independently, then

$$X = \frac{U}{U+V} \;\sim\; \text{Beta}(\alpha, \beta).$$

Algorithm:

1. Draw $U \sim \Gamma(\alpha, 1)$, i.e. sample

$$U \sim \text{Gamma}(\text{shape} = \alpha,\ \text{scale} = 1).$$

2. Draw $V \sim \Gamma(\beta, 1)$ independently.

3. Compute

$$X = \frac{U}{U + V}.$$

4. Return X, which follows $\text{Beta}(\alpha, \beta)$.

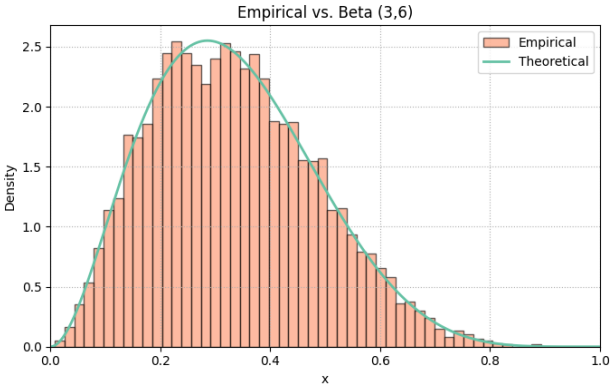


Figure 10: Beta$(3, 6)$ via Gamma ratio

3.11 Normal distribution

We wish to generate $Z \sim N(0, 1)$. Let $U_1, U_2 \overset{iid}{\sim} \text{Uniform}(0, 1)$. Define

$$R = \sqrt{-2\ln(U_1)}, \quad \Theta = 2\pi\, U_2,$$
$$Z_0 = R\cos(\Theta), \qquad Z_1 = R\sin(\Theta).$$

Then (Z_0, Z_1) has the joint density of two independent standard normals, since in polar coordinates the joint density

$$f_{R,\Theta}(r, \theta) = r \cdot \frac{1}{2\pi} \exp\left(-\tfrac{r^2}{2}\right)$$

transforms via the Jacobian r into

$$f_{Z_0,Z_1}(z_0, z_1) = \frac{1}{2\pi} \exp\Big(-\tfrac{z_0^2+z_1^2}{2}\Big),$$

the density of two independent $N(0,1)$ variables. Hence each of Z_0 and Z_1 is $N(0,1)$. (Figure 11)

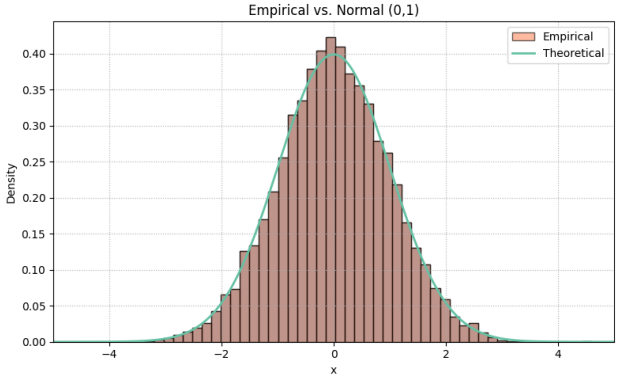


Figure 11: Box–Muller Generated Samples vs N(0,1) PDF

3.12 My own

Target distribution We want the uniform distribution on the right triangle (Fig 12)

$$\mathcal{T} = \{(x, y) : 0 < y < x < 1\}, \quad f_{X,Y}(x, y) = 2\, \mathbf{1}_{\mathcal{T}}(x, y).$$

Full conditionals Because the support is triangular,

$$X \mid Y = y \sim \text{Uniform}\big(y, 1\big), \qquad Y \mid X = x \sim \text{Uniform}\big(0, x\big).$$

Gibbs algorithm Given a starting point $(x^{(0)}, y^{(0)}) \in \mathcal{T}$, repeat for $t = 1, \ldots, T$:

$$x^{(t)} \leftarrow \text{Uniform}\big(y^{(t-1)}, 1\big),$$
$$y^{(t)} \leftarrow \text{Uniform}\big(0, x^{(t)}\big).$$

After discarding a burn-in of b steps, the draws $\{(x^{(t)}, y^{(t)})\}_{t>b}$ are samples from $f_{X,Y}$ (Figure 13).

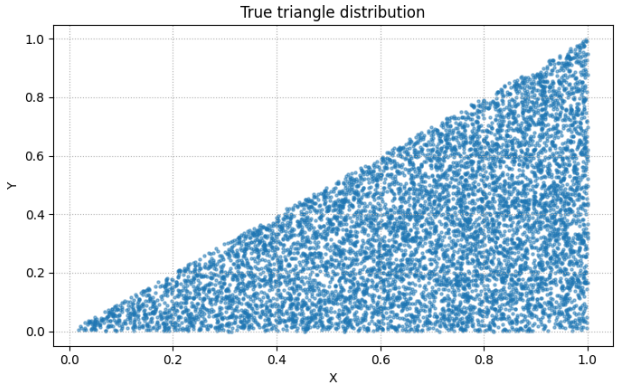


Figure 12

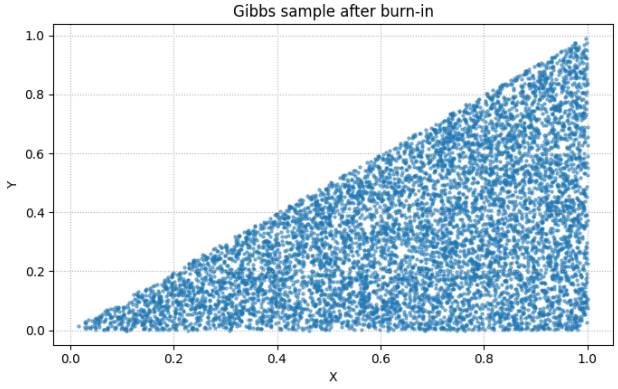


Figure 13

4 Conclusion

I dug into a bunch of the classic simulation tricks—Inverse Transform, Accept–Reject, sequential composition, the Box–Muller method, and Gibbs sampling—and actually coded them up to see how they perform. In every test, the histograms and scatterplots I got lined up almost perfectly with the theoretical PDFs or PMFs.

The inverse transform method is very useful in the easy distributions with clear CDFs, the accept-reject requires getting the best possible M, to increase the percentage of accepted samples, so the histograms align better with the real PDF. The Box – Muller method was

useful for the Normal distribution, while Gibbs method, eventhough it was at first hard to understand it, the code impementation was actually easy.

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